Class 8
Chemistry
Solutions

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1. ATOMIC STRUCTURE

Worksheet – 1

Level – I

1. A	4. A	7. C	10. B
2. B	5. C	8. C	
3. D	6. C	9. A	

Level – II

1) D	2) D	3) A	4) A	5) D	6) A	7) B	8) B
9) D	10) B	11) A	12) B	13) D	14) A	15) D	16) B
17) C	18) B	19) C	20) A,B,C,D		21) B	22) A	23)B
24) a-r, b-q, c-p, d-s		25) A,B,C	26) a-r, b-s, c-q, d-p		27) D	28) C	29) B
30) C							

Worksheet – 2

Level – I

1. D	4. B	7. B	10. A
2. A	5. A	8. A	
3. D	6. A	9. B	

Level – II

1) C	2) B	3) A	4) C	5) C	6) D	7) B	8) 2
9) A	10) A	11) A	12) D	13) C	14) A	15) D	16) 2
17) B	18) A	19) A	20) D	21) D	22) B	23) A,B,D	24) 2
25) A	26) a-s,t ;b-r;c-q;d-p		27) A,D	28) A,B,D	29) A,C,D		

Worksheet – 3

Level – I

1. A	4. A	7. A
2. C	5. B	8. B
3. A	6. A	9. A

Level – II

1) D	2) B	3) D	4) C	5) B	6) D	7) C	8) A
9) A	10) B	11) D	12) D	13) B	14) B	15) A	16) C
17) A	18) C	19) C	20) A	21) B,C	22) A	23) C	24) B
25) a-r ;b-q;c-s;d-p		26)A,B,C,D	27) A,B,C,D			28) B	29) A
30) B							

Worksheet – 4

Level – I

1. C
2. C
3. D
4. B
5. C
6. C
7. D
8. C
9. B
10. A

Level – II

1) A 2) C 3) A 4) C 5) B 6) C 7) B 8) B
9) A 10) A 11) C 12) D 13) A 14) B 15) A 16) B
17) B 18) D 19) C 20) C 21) C, D 22) D 23) A 24) B
25) a-s; b-r ;c-q; d-p 26) A, B, C, D 27) A, B 28) a-r b-p, s c-t d-q
29) 1, 2, 3, 4 30) a-w, b-v, c-u, d-t, e-s, f-r, g-q, h-p

1. $\lambda = \frac{c}{v} = \frac{3\times10^{10}}{1.0\times10^{6}} = 3 \times 10^{4}$ cm

2. $\lambda = 5000\ A^{0} = 5000 \times 10^{-8}$ cm

$c = 3 \times 10^{10}$ cm/s

$$v = \frac{c}{\lambda} = \frac{3 \times 10^{10}}{5 \times 10^{-5}} = 0.6 \times 10^{15} = 6 \times 10^{14}\ S^{-1}$$

4. $v = 80\text{MHz} = 80 \times 10^{6}$ Hz

$\lambda = ?\ \lambda = \frac{3\times10^{8}}{8\times10^{7}} = 3.75$ m

7. $\lambda = 720$ nm

$$v = \frac{c}{\lambda} = \frac{3 \times 10^{10}}{720 \times 10^{-7}} = 0.0416 \times 10^{16}$$

$= 4.16 \times 10^{14}\ S^{-1}$

9. $v = \frac{c}{\lambda} = \frac{3\times10^{8}}{400\times10^{-9}} = 0.75 \times 10^{15}$

$= 7.5 \times 10^{14}$ Hz

$$v = \frac{c}{\lambda} = \frac{3 \times 10^{8}}{750 \times 10^{-9}} = 0.04 \times 10^{16}$$

$= 4 \times 10^{14}$ Hz

10. $\lambda = \frac{c}{v} = \frac{3\times10^{8}}{1368\times10^{3}} = 0.00219 \times 10^{5}$

$= 219$ m

12. $\lambda = \frac{1}{\bar{v}} = \frac{1}{500}$

$$v = \frac{c}{\lambda} = c \times \bar{v}$$

$= 0.2 \times 10^{-2}$ cm

$= 3 \times 10^{10} \times 500$

$= 0.2 \times 10^{-2} \times 10^{-7} \times 10^{7}$ cm

$= 15 \times 10^{12}\ S^{-1}$

$= 2 \times 10^{4}$ nm

13. $\lambda = 5800\ A^0 = 5800 \times 10^{-8}$ cm

$\bar{v} = \frac{1}{\lambda} = \frac{1}{5800 \times 10^{-8}} = \frac{1}{58} \times 10^6 = 0.01724 \times 10^6 = 1.724 \times 10^4\ cm^{-1}$

$v = \frac{c}{\lambda} = \frac{3 \times 10^{10}}{5800 \times 10^{-8}} = \frac{3}{58} \times 10^{16} = 0.05172 \times 10^{16} = 5.172 \times 10^{14}\ S^{-1}$

14. $\vartheta = 4.69 \times 10^{14} Sec^{-1}$

$\vartheta = \frac{C}{\lambda}$

$c = 3 \times 10^8$ m

$\lambda = \frac{c}{\vartheta} = \frac{3 \times 10^8}{4.69 \times 10^{14}}$

$= \frac{3}{4.69} \times 10^8 \times 10^{-14}$

$= 0.639 \times 10^{-6}$

$= 0.64 \times 10^{-6}; = 6400 \times 10^{-10}$ m; $\lambda = 6400\ A^0$

15. $E \propto V$

$c = 340\ S^{-1};\ \vartheta = \frac{C}{\lambda};\ \lambda = \frac{c}{\vartheta};\ \lambda = \frac{340}{700}$

$\lambda = 0.485$

16. $C = 3 \times 10^8$ m/sec

$\vartheta = 12 \times 10^{16}$ Hz; $\vartheta = \frac{C}{\lambda}; \lambda = \frac{C}{v} = \frac{3 \times 10^8}{12 \times 10^{16}}; = 0.25 \times 10^{-8}$ m

$= 2.5 \times 10^{-9}$ m

17. $\lambda = 6000\ A^0$

$v = \frac{c}{\lambda}; = \frac{3 \times 10^8\ m/s}{6000 \times 10^{-10}\ m} = \frac{1}{2 \times 10^{-15}} sec^{-1}$

$= 5 \times 10^{14} sec^{-1}$

18. $\vartheta = \frac{c}{\lambda}$

$\vartheta = c \times \frac{1}{\lambda}; \vartheta = c \times \bar{o}; \vartheta = 3 \times 10^{10} \times 200 = 6 \times 10^{12}\ S^{-1}$

20. Wavelength $\lambda_g = ?$

$\vartheta = \frac{c}{\lambda}; \lambda_g = \frac{c}{v}; \lambda_g = \frac{3 \times 10^8\ m/x}{5.75 \times 10^{14} H_2}; \lambda_g = \frac{3}{5.75} \times 10^8 \times 10^{-14}$

$= \frac{3}{5.75} \times 10^{-6}$ m; $= 0.521 \times 10^{-6} \times \frac{10^3}{10^3}; = 521 \times 10^{-9}$ m; $= 521$ nm

$\lambda_y = \frac{c}{\vartheta} = \frac{3 \times 10^8}{5.15 \times 10^{14}}; = \frac{3}{5.15} \times 10^8 \times 10^{-14}; = \frac{3}{5.15} \times 10^{-6}; = 0.582 \times 10^{-6} \times \frac{10^3}{10^3}$

$= 582 \times 10^{-9}$ m; $\lambda_R = 582$ nm; $R = \frac{c}{\vartheta} = \frac{3 \times 10^8}{4.27 \times 10^{14}}$

$= \frac{3}{4.27} \times 10^{-6}; = 0.702 \times 10^{-6} \times \frac{10^3}{10^3}; \lambda_R = 702 \times 10^{-9}; = 702$ nm

24. $\bar{v} = 97540\ cm^{-1}; \bar{v} = c\bar{v}; = 3 \times 10^{10}\ cmsec^{-1} \times 97540\ cm^{-1}$

$= 2.926 \times 10^{15}\ s^{-1}$

Worksheet – 5

Level – I

1. D
2. C
3. D
4. D
5. B
6. D
7. A
8. A
9. C
10. A

Level – II

1) D 2) C 3) D 4) B 5) C 6) C 7) 4 8) 1
9) D 10) A 11) D 12) C 13)A 14) C 15) 1 16) 2
17) A 18) C 19) D 20) A,D 21) A,B,C 22) A
23) a-s, b-q, c-r, d-p 24) 3 25) D 26) B 27) B, C, D 28) C 29) C
30) C 31) a-r, s; b-p;c-t ;d-q 32) a-s; b-r, s; c-q,t; d-p

1. J-s or Erg-s
2. $E = 1\ J$

   Photon wave number $= x = \frac{1}{\lambda}$

   $\Rightarrow \lambda = \frac{1}{x}$

   $\Rightarrow$ Energy of 1 photon = hcxJ

   $\therefore$ No. of photons in 1 J source $= \frac{1}{hcx}$
4. $\Rightarrow w_0 = 3 \cdot 3eV = hv$

   $\Rightarrow v_0 = \frac{3.3eV}{h}$

   $v_0 = \frac{3.3 \times \frac{1}{167 \times 10^{-19}}\ J}{6.626 \times 10^{-34}\ J.S} = 7.96 \times 10^{14}\ Hz$
5. $E = \frac{1}{2}mV^2 + W$ or $E = KE + W$
6. $\frac{\lambda_1}{\lambda_2} = \frac{E_1}{E_2}; \frac{600\ nm}{300\ nm} = \frac{E_1}{E_2}; \frac{2}{1} = \frac{E_1}{E_2}$
7. $hv = w + KE; K.E = hv - w; = 6.2 - 1.2; = 5ev$
8. $k = nh\frac{c}{\lambda}; = \frac{5.6 \times 10^{24} \times 6.626 \times 10^{-34} \times 3 \times 10^8}{337.1 \times 10^{-9}}$

   $= 3.33 \times 10^6\ J$
11. $\frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{4000}{2000} = 2$
12. $W_O = E - KE; E = 5.6 \times 10^{-19}\ J; K.E = 12.0 \times 10^{20}\ J; E = W^0 + K \cdot E$

    $= 5.6 \times 10^{-19}\ J - 12.0 \times 10^{-20}\ J$

    $= (5.6 - 1.2) \times 10^{-19}$

    $= 4.4 \times 10^{-19}\ J$
13. $E \propto v; \frac{E_1}{E_2} = \frac{v_1}{v_2}; 14. E = hc\bar{v}$

    $E = 6.624 \times 10^{-34} \times 3 \times 10^8 \times 5 \times 10^{-5}$

    $= 9.93 \times 10^{-30}\ J$
15. $E = hv$

    $E = 6.624 \times 10^{-34} \times 3 \times 10^{15}\ s^{-1}$

    $= 1.99 \times 10^{-B} J$

16. $E = nhu$

$n = \frac{E}{hu}$

17. $E = \frac{hc}{\lambda}$

18. $E = hc\bar{U}$

19. $\Rightarrow KE_1 : KE_2 = 1 : K; \Rightarrow hv_1 = KE_1 + hv_0; \Rightarrow hv_2 = KE_2 + hv_0$

$\Rightarrow KE_1 = h(v_1 - v_0)$

$KE_2 = h(v_2 - v_0)$

$\Rightarrow \frac{KE_1}{KE_2} = \frac{1}{K} = \frac{h(v_1 - v_0)}{h(v_2 - v_0)}$

$(v_2 - v_0) = K(v_1 - v_0)$

$Kv_0 - v_0 = Kv_1 - v_2$

$\Rightarrow v = \frac{kv_1 - v_2}{k - 1}$

Worksheet – 6

Level – I

1. A	4. A	7. D	10. B
2. D	5. C	8. A	
3. A	6. A	9. A	

Level – II

1) 4	2) 2	3) 2	4) 2	5) 2	6) 2	7) 1	8) 3
9) 4	10) 3	11) 1	12) 3	13) 4	14) 4	15) 3	16) 1
17) 3	18) 1, 2, 3, 4		19) 1, 2, 3, 4		20) 1	21) 4	22) 4
23) a-q,b-s,c-r,d-p		24) 2,3	25) a-q ;b-p ; c-r ;d-s		26) 1	27) 4	
28) a- t ;b-p, r,; c-p,q ;d-s							

4. When electron jumps from n^{th} orbit to ground state

$$\frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

11. For H_a line $n_1 = 2, n_2 = 3$

$$\therefore \bar{v} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$= R\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = R\left[\frac{1}{4} - \frac{1}{9}\right] = \frac{5R}{36}$$

Worksheet – 7

Level – I

1. B	6. B	11. B	16. A
2. C	7. A	12. A	17. A
3. D	8. B	13. A	
4. C	9. A	14. D	
5. B	10. D	15. D	

Level – II

1) D	2) A	3) D	4) A	5) 1	6) 3	7) 4	8) 2
9) A	10) A	11) D	12) C	13) 1	14) 4	15) 1	16) 1
17) D	18) A,B,C	19) C	20) D	21) 3	22) a-p,r; b-r,t; c-s; d-q		
23) A,C,D	24) B	25) A,B,C	26) C	27) 4	28) 3		
29) a-p,t; b-p; c-p,r,s; d-p,q			30) D				

4. $E_n = -\frac{13.6}{n^2}eV, E_5 = -\frac{13.6}{5^2} = -\frac{13.6}{25} = -0.544eV$

5. $r = 0.529 \times \frac{n^2}{Z}$; $= 0.529 \times 16 = 8.464$ A°

6. $r = 0.529 \times \frac{n^2}{Z}$; $= 0.529 \times 100 = 52.9$ A°

7. $r = 0.529 \times \frac{n^2}{Z}$; Ratio: $1 : \frac{1}{2} : \frac{1}{3}$
$6 : 3 :$

8. $r = \frac{0.52\times10^{-8}\times1^2}{2} = 0.26 \times 10^{-8}$ cm

9. $E_n = -\frac{13.6}{n^2}Z^2eV$

$(I.P.)_H = E_\infty - E_1 = 13.6eV; (I.P.)_{He+} = E_\infty - E_1 = 13.6 \times 2^2 = 54.4eV$

13. $r_n = \frac{n^2h^2}{4\pi^2mze^2} \times \frac{1}{4\pi\epsilon_0}$

14. Energy liberated $= E_3 - E_1;; E_n = -\frac{21.76}{n^2} \times 10^{-12}$ergs

$\Delta E = E_3 - E_1 = 21.76 \times 10^{-12}\left(\frac{1}{3^2} - \frac{1}{1^2}\right)$ ergs $; = 21.76 \times 10^{-12} \times \frac{8}{9} = 19.34 \times 10^{-12}$

$= 0.1934 \times 10^{-10}$ ergs.

17. $\frac{P \cdot E}{T \cdot E} = 2$

Worksheet – 8

Level – I

1) C	2) D	3) B	4) C	5) C	6) D	7) D
8) A	9) C	10) B				

Level – II

1) 4	2) 3	3) 2	4) 1	5) 2	6) 3	7) 4
8) 4	9) 2	10) 4	11) 3	12) 1	13) 1	14) 3
15) 1	16) 3	17) 3	18) 1,2,3	19) 1,3,4	20) 4	21) 4
22) 4	23) a-r b-q c-s d-p		24) 1,2,3,4	25) 1,2		26) a-r ,b-s,t ,c-p d-q
27) 1,2,3,4	28) 4					

2. $m = 0.2$ kg

$V = 10\ m \cdot s^{-1}$

$\lambda = \frac{h}{mv}$

$= \frac{6.626 \times 10^{-34}}{0.2 \times 10^{-34} \times 10^{-1}}$

$= 33.13 \times 10^{-34}$ m

$= 3.313 \times 10^{-34}$

4. $\Delta x \cdot m\Delta v \geq \frac{h}{4\pi}; \Delta x \geq \frac{h}{4\pi m\Delta v}$

$\geq \frac{6.626 \times 10^{-27}}{4 \times 3.14 \times 9.1 \times 10^{-28} \times 2 \times 10^{-3}}; = 0.2858 \times 10^4 = 2.9 \times 10^2$ cm

7. $v = 1 \times 10^{-3}$ m/s; $m = 1.67 \times 10^{-27}$ kg

$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}}{1.67 \times 10^{-27} \times 1 \times 10^{-3}}$

$= 3.967 \times 10^{-4}$ m; $= 3.967 \times 10^{-9}$ m $= 0.4$ nm

10. $\lambda = \frac{h}{\sqrt{2eVm}} = \frac{\lambda_{Be^{+3}}}{\lambda_{proton}} = \sqrt{\frac{m_P}{m_{Be^{+3}}}} = \sqrt{\frac{1}{9}}$

$\lambda_{Be^{+3}} : \lambda_{proton} = 1 : 3\sqrt{3}$

11. $V = 200$ m/s

$\Delta V = 1$ m/s

$\Delta x = ?$

$\Delta x \cdot m\Delta V \geq \frac{h}{4\pi}$

$\Delta x \geq \frac{h}{4\pi m\Delta V}$

$= \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 1} = 0.058 \times 10^{-3} = 58$

12. $\Delta x = \frac{h}{4\pi m\Delta V}$

$= \frac{6.626 \times 10^{-27}}{4 \times 3.14 \times 9.1 \times 10^{-28} \times 2 \times 10^{-3}} = 0.0289 \times 10^4$

$= 2.89 \times 10^2$ cm

13. $\lambda = 1.0 \times 10^2$ nm

$\frac{\lambda}{1} = \frac{h}{mV}; V = \frac{h}{m\lambda} = \frac{6.626 \times 10^{-27}}{9.1 \times 10^{-28} \times 1 \times 10^2 \times 10^{-7}}$

$= 0.7281 \times 10^6$ cm/s $= 7.2 \times 10^5$ cm/s

14. $\lambda = \frac{6.626\times10^{-34}}{0.5\times100} = \frac{6.626}{5} \times 10^{-34} \times 10^{-1}$
$= 1.32 \times 10^{-35}$ m

16. $\lambda = \frac{h}{mv}; = \frac{6.626\times10^{-34}}{3.3\times10^{-24}} = 2\ A^0$

17. $\lambda = \frac{h}{mv}; m = \frac{h}{\lambda v}; = \frac{6.625\times10^{-34}}{3.6\times10^{-10}\times3\times10^{8}}; = 6.1 \times 10^{-33}$ kg

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Worksheet – 9

Level – I

1. D
2. B
3. A
4. C
5. 2
6. 1
7. 1
8. 1
9. 3
10. 2

Level – II

1) 2 2) 3 3) 1 4) 2 5) 4 6) 2 7) 2 8) 4
9) 1 10) 4 11) 2 12) 3 13) 1 14) 2 15) 4 16) 3
17) 4 18) 1,2,3,4 19) 2,3 20) 3 21) a-p b-q c-r d-s
22) 3,4 23) 1,2,3 24) 1,2,3 25) a-s b-p c-q d-r 26) 3 27) 4 28) 4
29) a-q , b-r , c-p, d-s

Worksheet – 10

Level – I

1. B
2. B
3. B
4. A
5. B
6. B
7. C
8. D
9. C
10. D

Level – II

1) A 2) B 3) D 4) C 5) D 6) C 7) A 8) C
9) C 10) A 11) C 12) C 13) C 14) A 15) B 16) A
17) A,B 18) B,C 19) B,C,D 20) C 21) B 22) B
23) a-s b-r c-p d-q,t 24) a-p,r, s: b-r,s, t c-q,r,s: d-p, r

2. Periodic Classifications

Worksheet – 1

Level – I

1. C
2. B
3. A
4. A
5. B
6. C
7. D
8. A
9. B
10. B

Level – II

1) B 2) A 3) B 4) D 5) A 6) C 7) B 8) D
9) D 10) B 11) C 12) C 13) B 14) A 15) A 16) A
17) A 18) D 19) A 20) C,D 21) B,C 22) D 23) D 24)C
25) a-p ; b-q,r ; c-s ;d-t 26) C,D 27) A,B,C 28) a-q; b-r ; c-r ;d-s
29) A,B,C 30) a-p ; b-r ; c-s ;d-q

Worksheet – 2

Level – I

1. A
2. B
3. C
4. D
5. C
6. A
7. B
8. B
9. A
10. A
11. A

Level – II

1) 2 2) 3 3) 1 4) 2 5) 2 6) 2 7) 2 8) 3
9) 3 10) 3 11)2 12) 3 13) 3 14) 4 15) 3 16) 3
17) 4 18) 1 19) 1 20) 1,2,3 21) 1,2,3,4 22) 2 23) 3 24) 1
25) a-s,t ; b-r ; c-q; d-p 26) 3 27) 3 28) 3
29) a-s ; b-t ;c-q; d-p e-r; 30) 4 31) a-r ; b-s ; c-q; d-p

Worksheet – 3

Level – I

1) A 2) D 3) A 4) A 5) B 6) D 7) D

Level – II

1) B 2) A 3) B 4) C 5) C 6) C 7) A 8) A
9) C 10) B 11) B 12) C 13) D 14) C 15) C 16) A
17) A 18) A,B 19) D 20) B 21) A 22) a-s ; b-q ; c-r ;d-p
23) C 24) C 25) B 26) A 27) B 28) C
29) a-s ; b-r ; c-q ;d-p

Worksheet – 4

Level – I

1) C 2) C 3) B 4) C 5) A 6) A

Level – II

1) C	2) B	3) B	4) A	5) C	6) C	7) D	8) B
9) B	10) A	11) A	12) D	13) B	14) B	15) C	16) A
17) C	18) B	19) D	20) D	21) B	22) A,C,D	23) C	24) A
25) B	26) a-q,t ; b-s ; c-p ;d-r			27) A	28) B,C		
29) a-q ; b-s ;c-p ;d r		30) A	31) A	32) a-q ; b-p,r,s,t ; c-p ;d-q			

Worksheet – 5

Level – I

1) D	2) D	3) B	4) A	5) B	6) D	7) D	8) D
9) C	10) A	11) D					

Level – II

1) D	2) A	3) A	4) A	5) A	6) B	7) C	8) A
9) D	10) A	11) D	12) B	13) B	14) C	15) A	16) D
17) D	18) A,B,C,D	19) B	20) A	21) C	22) a-t ; b-s ; c-p,q ;d-r		
23) A,B	24) A,B,C	25) D	26) A,C	27) C,D	28) B	29) B	30) B

Worksheet – 6

Level – I

1) A	2) B	3) A	4) A	5) A	6) C	7) A	8) A

Level – II

1) B	2) A	3) C	4) B	5) A	6) D	7) B	8) B
9) C	10) C	11) C	12) C	13) A	14) C	15) A	16) D
17) A	18) A	19) C	20) A	21) A,B,C	22) D	23) A	24) A
25) a-p ; b-q ; c-r ;d-s		26) A,B,C	27) B,C,D	28) A	29) C,D	30) A,B,C	

Worksheet – 7

Level – I

1) B	2) B	3) C	4) D	5) C	6) B	7) D	8) B
9) C	10) B	11) A	12) B	13) A			

Level – II

1) D	2) B	3) A	4) C	5) B	6) D	7) D	8) B
9) C	10) A	11) D	12) C	13) C	14) D	15) B	16) C
17) C	18) D	19) D	20) C	21) C	22) A,B	23) A,B,C	
24) A,B,C,D		25) A,B	26) A,B	27) B	28) A	29) C	30) D
31) a-s; b-r; c-p;d-q		32) B	33) a-s; b-q; c-r;d-p		34) a-s; b-p; c-q;d-r		

3. Mole Concept and Stoichiometry

Worksheet – 1

Level – I

1. B	3. D	5. B	7. D
2. C	4. A	6. B	

Level – II

1) C	2) A	3) A	4) C	5) A	6) B	7) D	8) A
9) B	10) B	11) A	12) C	13) C	14) A	15) C	16) C
17) B,C	18) B, C	19) D	20) C	21) B	22) a-p; b-p, c-q, d-r		
23) A, B, C	24) A, C, D	25) B	26) A	27) A	28) a-r; b-t, c-p, d-q		

1. Weight of the atom = 4 grams = $4 \times 1.66 \times 10^{-24} = 6.64 \times 10^{-24}$
2. Gram atoms = $\frac{8}{16} = 0.5$
3. Weight = number of gram atoms × atomic weight = $4 \times 14 = 56$ g
4. Volume = n × 2.24 = $\frac{4.4}{44} \times 22.4 = 2.24$ lit
5. Number of Gram molecules = $\frac{9.8}{98} = 0.1$
6. a) Number of Gram molecules = $\frac{22}{44} = 0.5$
   b) Number of Gram molecules = $\frac{49}{98} = 0.5$
   c) Number of Gram molecules = $\frac{53}{106} = 0.5$
   d) Number of Gram molecules = $\frac{10}{100} = 0.1$
11. Number of Gram atoms = $\frac{120}{24} = 5$
12. 6.023×10^{23} molecules of N_2 weight = 2.8 grams
    3.01×10^{23} molecules of SO_2 weight = 32 grams
    Total weight = 60 grams
14. 6.023×10^{23} atom of Hydrogen weight = 1 gram
    3.01×10^{23} molecular of Hydrogen weight = 2 grams ratio = 1: 2

a. Weight of 5 gram atoms of Nitrogen = 5×14

3. 1 gr - atom of silver weight = $1 \times 108 = 108$ grams
4. 5×10^{23} atom of Carbon ⇒ $\frac{5 \times 10^{23} \times 12}{6 \times 10^{23}} = 10$ grams
16. The number of gram atoms = $\frac{0.004}{24} \times N = \frac{1}{6} \times 10^{-3} \times 6 \times 10^{23} = 10^{20}$
17. 2800 grams (or) $28 \times 10^2 \times 1.66 \times 10^{-24} = 28 \times 1.66 \times 10^{-22}$gr
18. If atomic weight of Oxygen is 100 , then Atomic weight of Hydrogen = $\frac{1 \times 100}{16} = 6.25$

Mass of $H_2O = 2(6.25) + 100 = 112.5$

24. Weight of NH_3 molecule = 17 grams (or) $17 \times 1.66 \times 10^{-24}$ g (or) $17 \times 1.66 \times 10^{-27}$ kg
25. 1 gram molecule of $CH_4 \rightarrow 2$ gram molecule of H_2
26. x = 64 grams, Number of gram molecules = $\frac{64}{16} = 4$

Worksheet – 2

Level – I

1. D
2. B
3. D
4. B
5. D
6. A
7. B

Level – II

1) A 2) C 3) D 4) A 5) A 6) B
7) A 8) C 9) A 10) A 11) A 12) B
13) D 14) C 15) A, B, C, D 16) A, C, D 17) A 18) B
19) C 20) a-p,q; b-r, c-q, d-s 21) A 22) A, B, D
23) a-r; b-s, c-p, d-q 24) A, C, D 25) A 26) B 27) A
28) a-s; b-p, c-t, d-q

3. Number of atom in 12 grams of Mg $= \frac{12}{24}N_A = \frac{1}{2}N_A$
   Number of atom in 7 grams of CO $= \frac{7}{28}N_A \times 2 = \frac{1}{2}N_A$
4. Number of molecules $= \frac{44}{44} \times 6 \times 10^{23} = 6 \times 10^{23}$
5. Number of atoms $0.1 \times N_A \times 3 = 0.3 \times 6.023 \times 10^{23} = 1.806 \times 10^{23}$
6. Number of Cl^-ion $= \frac{222}{111} \times N_A \times 2 = 2N_A$
7. Number of Oxygen atoms $= \frac{10.6}{106} \times 6.023 \times 10^{23} \times 3 = 1.806 \times 10^{23}$
8. $6.02 \times 10^{23} \rightarrow 1$ mole
   $6.02 \times 10^{25} \rightarrow ?; \frac{6.02\times10^{25}\times1}{6.02\times10^{23}} = 10^2 = 100$ moles
   $\therefore$ Number of gram-molecule of $Cl_2 = \frac{100}{2} = 50$
9. 16 g of $O_2 \rightarrow \frac{16}{32} = 0.5$ moles
   14 g of $N_2 \rightarrow \frac{14}{28} = 0.5$moles
10. No. of gram-molecules $= \frac{13.5}{135} = 0.1$
11. 10^{21} molecular of CO_2 means
    $\Rightarrow \frac{10^{21} \times 44}{6 \times 10^{23}}$
    $\Rightarrow 7.33 \times 10^{-2}$ g
    $\Rightarrow$ 73.3mg
    weight of CO_2 lift $= 200 - 73.3$
    $= 126.6$mg
    moles $= \frac{126.6\times10^{-3}}{44} = 2.87 \times 10^{-3}$
12. 10 lac
    $\rightarrow$ per Second
    per minute $\rightarrow 60 \times 10$ lac
    per hour $\rightarrow 60 \times 60 \times 10$ lac
    per day $\rightarrow 24 \times 60 \times 60 \times 10$ lac
    per year $\rightarrow 365 \times 24 \times 60 \times 60 \times 10$ lac
    31.5×10^7 lac per year
    $31.5 \times 10^7 \times 10^5 \rightarrow 1$ year
    $6 \times 10^{23} \rightarrow ?$

$\frac{6 \times 10^{23}}{31.5 \times 10^{7} \times 105} = 0.19 \times 10^{11}$

1.9×10^{10} year

13. Molecular weight of $P_2Q_3 = 2a + 3\ b$

Molecular weight of $PQ_2 = a + 2\ b$

given that 0.15 mole of P_2Q_3 weight 15.9gm

$(2a + 3b) = \frac{15.9}{0.15}$

$(a + 2b) = \frac{9.3}{0.15}$

solving these two equations

$a = 26, b = 18$

14. wt of water = 0.01gm

wt of $D_2O \Rightarrow$?

$6000 \rightarrow 1$ part

$0.01 \rightarrow ? \frac{0.01}{6000} = 0.167 \times 10^{-5}$

No. of molecules $= \frac{0.167 \times 10^{-5}}{20} \times 6.023 \times 10^{23}$

$= 0.05 \times 10^{18}$

$= 5 \times 10^{16}$

16. Molecule $= \frac{11}{44} N = \frac{1}{4}(N)$

1. 8 g of $O_2 \Rightarrow \frac{8}{32} N = \frac{1}{4} N$

2. 16 g of $O_2 \Rightarrow \frac{16}{32}$ N $= \frac{1}{2}$ N

3. 7 g of $CO \Rightarrow \frac{7}{28}$ N $= \frac{1}{4}$ N

4. 7 g of $N_2 \Rightarrow \frac{7}{28} N = \frac{1}{4} N$

17. $6 \times 10^{23} \rightarrow 40$

$2 \times 10^{24} \rightarrow ? : \frac{2 \times 10^{24} \times 40}{6 \times 10^{23}} = 13.33 \times 10 = 133.3$

19. $6.023 \times 10^{23} \rightarrow 40$

$3 \times 10^{24} \rightarrow ?$

$\frac{3 \times 10^{24} \times 40}{6 \times 10^{23}} = 20 \times 10 = 200$

249.8 grams of H_2SO_4

No. of moles of $H_2SO_4 = \frac{9.8}{98} = 0.1$

Worksheet – 3

Level – I

1. A	3. B	5. A	7. C
2. A	4. B	6. C	8. D

Level – II

1. B	3. C	5. B	7. A
2. A	4. D	6. B	8. B

9. B
10. B
11. D
12. D
13. B
14. A, B, D
15. B,D
16. B
17. A
18. A
19. a-s; b-q, c-p, d-r
20. 0
21. A
22. A
23. C
24. A, B, C
25. B
26. D
27. A
28. a-p,r; b-q,r, c-p,s, d-q,t

5 (i) $K_2Cr_2O_7$

$2(-4) + 2x + 7(-2) = 0$

$x = +6$

(ii) $K_2H_2Sb_2O_7$

$+2 + 2 + 2x - 14 = 0; 2x = 10; x = +5$

6. $C_3O_2; 3x + 2(-2) = 0; x = \frac{+4}{3}$
7. $K_4[Fe(CN)_6]: +4 + x + 6(-1) = 0; x = +2$
8. $K_2MnO_4: +2 + x + 4(-2) = 0; +x - 6 = 0; x = +6$
9. $C_6H_{12}O_6 \rightarrow 6x + 12 - 12 = 0; x = 0$

$CH_3OH \rightarrow x + 4 - 2 = 0; x = -2;$

$NH_4^{+1} \rightarrow x + 4(+1) = +1; x = -3$

a. $SO_3 \rightarrow x + 3(-2) = 0; x = +6;$
2. $SO_4^{-2} \rightarrow x + 4(-2) = -2; x = +6$
3. $SO_3^{-2} \rightarrow x + 3(-2) = -2; x - 6 = -2;$
4. $S_2O_8^{-2} \rightarrow 2x + 8(-2) = -2; 2x = +14$

$x = +7$ should not exceed group number, then oxidation number = +6

15. $CaOCl_2$
16. $CaH_2 \rightarrow +2 + 2x = 0; 2x = -2; x = -1$
17. $NH_4NO_2 \rightarrow NH_4^{+1}, NO_2^{-1}$

$x + 4 = +1x + 2(-2) = -1; x = -3x = +3$

Worksheet – 4

Level – I

1. B
2. A
3. D
4. D
5. D
6. B
7. D

Level – II

1) A	2) C	3) D	4) A	5) C	6) A	7) A	8) B
9) C	10) D	11) D	12) A	13) A	14) A, C	15) A, C	16) C
17) C	18) C	19) a-r; b-s; c-q, d-p		20) B, D	21) A	22) B	23) A

24) a-r,t; b-s, c-p, d-q 25) A 26) C 27) C 28) a-r; b-p; c-s,: d-q

Worksheet – 5

Level – I

1. C 3. C 5. C 7. B
2. C 4. A 6. D 8. A

Level – II

1) 4 2) 3 3) 1 4) 4 5) 4 6) 1 7) 1 8) 4
9) 4 10) 2 11) 1 12) 3 13) 1, 3 14) 1 15) 3 16) 2
17) a-t; b-r, c-q, d-p 18) 1 19) 3 20) 1 21) 4
22) a-r; b-s,t; c-p; d-q,t 23) 1, 4 24) 2, 3 25) 2 26) 3 27) 4
28) a-r; b-p; c-s,t; d-q

Worksheet – 6

Level – I

1. C 3. A 5. D 7. B
2. B 4. B 6. A

Level – II

1. C 7. B 13. C 19. C
2. C 8. B 14. C 20. D
3. B 9. A 15. A 21. A
4. D 10. B 16. B
5. B 11. A 17. C
6. D 12. C 18. A,C

22. a-q; b-p; c-p; d-p,t

3. $2MnO_4^- + 5H_2O_2 + \underset{\downarrow}{x}\,H^+ \rightarrow 2Mn^{+2} + 8H_2O + 5O_2$

4. $Zn + NO_3^- \rightarrow Zn^{+2} + NH_4^+$

$Zn \rightarrow Zn^{+2}$ $NO_3^- \rightarrow NH_4^+$

$Zn \rightarrow Zn^{+2} + 2e^-$ $NO_3^- \rightarrow NH_4^+ + 3H_2O$

$4Zn \rightarrow 4Zn^{+2} + 8e^-$ $NO_3^- + 10H_2O \rightarrow NH_4^+ + 10OH^- + 3H_2O$

$NO_3^- + 10H_2O \rightarrow NH_4^+ + 10OH^- + 3H_2O + 8e^-$

$NO_3^- + 10H_2O + 8e^- \rightarrow NH_4^- + 10OH^- + 3H_2O$

$Zn + NO_3^- + 7H_2O \rightarrow 4Zn^{+2} + NH_4^+ + 10O_4^-$

6. $BiO_3^- + 6H^+ + xe^- \rightarrow Bi^{+3} + 3H_2O$

$BiO_3^- + 6H^+ + 2e^- \rightarrow Bi^{+3} + 3H_2O$

8. $2IO_3^- + 5SO_2 + 4H_2O \rightarrow I_2 + 5SO_4^{-2} + 8H^+$

9. Number of electrons transferred $= 1 \times 5e^-$

10. $Cr_2O_7^{-2} + 14H^+ + XI^- \rightarrow 2Cr^{+3} + 7H_2 + 8H^+$

$Cr_2O_7^{-2} + 14H^+ + 6I^- \rightarrow 2Cr^{+3} + 7H_2O + 3I_2$

13. $Cr_2O_7^{-2} + 14H^+ + 6Fe^{+2} \rightarrow 2Cr^{+3} + 6Fe^{+3} + 7H_2O$

14. The balanced chemical equation is $14H^+ + Cr_2O_7^{-2} + 4I^- \rightarrow 2Cr^{+3} + 3I_2 + 7H_2O$
Thus, 3 moles of I_2 is evolved, when KI reacts with 1 mole of Potassium dichromate.

15. $MnO_4^- + H_2O_2 \rightarrow Mn^{+2} + H_2O + O_2$

$1X5=5e^-$

$2MnO_4^- + 5H_2O_2 \rightarrow 2Mn^{+2} + 8H_2O + 5O_2 + 6H^+$
2 mole of $KMnO_4 \rightarrow 5$ moles of H_2O_2
1 mole of $KMnO_4 \rightarrow$?
$\frac{1X5}{2} = \frac{5}{2}$ mole of H_2O_2

19. $Al + 2H_2O \rightarrow AlO_2^- + 4H^+ + 3e^-$

a. $C_2H_2 + MnO_4^- \rightarrow CO_2 + Mn^{+2}$

Oxidation Half reaction	Reduction Half Reaction
$C_2H_2 \rightarrow CO_2$	$MnO_4^- \rightarrow Mn^{+2}$
$C_2H_2 \rightarrow 2CO_2$	$MnO_4^- \rightarrow Mn^{+2} + 4H_2O$
$C_2H_2 + 4H_2O \rightarrow 2CO_2$	$MnO_4^- + 8H^+ \rightarrow Mn^{+2} + 4H_2O$
$C_2H_2 + 4H_2O \rightarrow 2CO_2 + 8H^+$	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O$
$C_2H_2 + 4H_2O \rightarrow 2CO_2 + 8H^+ + 10e^-$	$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O + 8H_2O$
$\therefore C_2H_2 + 4H_2O \rightarrow 2CO_2 + 8H^+ + 10e^-$	$\therefore 2MnO_4^- + 16H^+ + 10e^- \rightarrow 2Mn^{+2} + 8H_2O$
$C_2H_2 + 2MnO_4^- + 6H^+ \rightarrow 2CO_2 + 2Mn^{+2} + 4H_2O$	

$8e^- \rightarrow 4e^-$

23. $Zn + NO_3^- \xrightarrow{OH^-} NH_4^+ + Zn^{+2}$

$2e^- \rightarrow 1e^-$

$4-5 + NO_3^- \rightarrow NH_4^+ + Zn^{+2}$
$4Zn + NO_3^- \rightarrow NH_4^+ + 4Zn^{+2} + 3H_2O$
$4Zn + NO_3^- + 10H_2O \rightarrow NH_4^+ + 4Zn^{+2} + 3H_2O + 10OH^-$
$4Zn + NO_3^- + 7H_2O \rightarrow NH_4^+ + 4Zn^{+2} + 10OH^-$

24. $AsO_3^{-3} + MnO_4^- \longrightarrow AsO_4^{-3} + MnO_2$

$2e^-$

$$\overset{-3}{\underset{MnOAO_4^-}{\overset{+2H_2O}{}}} \rightarrow 3AsO_4^{-3} + 2MnO_2 + \underset{+2OH_2^-}{H_2O}$$

$$3AsO_3^{-3} + 2MnO_4^- \rightarrow 3AsO_4^{-3} + 2MnO_2 + H_2O$$

Worksheet – 7

Level – I

1) B	2) A	3) C	4) C	5) C	6) B	7) D
8) A	9) C	10) A	11) C			

Level – II

1) C	2) A	3) A	4) A	5) A	6) B	7) A
8) A	9) B	10) D	11) B	12) A	13) A	14) A
15) D	16) B	17) D	18) D	19) A, B	20) A	21) C
22) D	23) a-s; b-p; c-q; d-r		24) C	25) C	26) D	27) B
28) A, B, C						

1. % of C $= \frac{12}{16} \times 100 = 75\%$
2. % of O $= \frac{16}{18} \times 100 = 100 = 88.89\%$
3. % of $H_2O = \frac{2\times18}{172} \times 100 = 20.93\%$
4. % of C $= \frac{72}{78} \times 100 = 92.3\%$
5. % of C $= \frac{12}{44} \times 100 = 27.27\%$
6. $40 = \frac{x\times12}{90} \times 100 = x = 3$
7. $0.33 = \frac{2\times56}{x} \times 100$

   $X = \frac{2 \times 56}{0.33} \times 100 = 33939$

10. % of N $= \frac{28}{22400} \times \frac{50}{0.132} \times 100 = 47.34\%$
11. i) $C_6H_6 \rightarrow$ % of C $= \frac{72}{78} \times 100 = 92.30$

    (ii) $(CH_3)_2CO \rightarrow$ % of C $= \frac{36}{58} \times 100 = 62.06\%$

    (iii) $CH_3O_4 \rightarrow$ % of C $= \frac{12}{79} \times 100 = 37.5$

    (iv) $CCl_4 \rightarrow$ % of C $= \frac{12}{154} \times 100 = 7.79$
12. (i) % of Ag $= \frac{108}{143.5} \times 100 = 75.26\%$

    (ii) % of Fe $= \frac{2\times56}{106} \times 100 = 70\%$

    (iii) % of K $= \frac{78}{174} \times 100 = 44.82\%$
13. % of N $= \frac{28}{22400} \times \frac{112}{0.5} \times 100 = 28\%$
14. % of C $= \frac{12}{44} \times \frac{1.32}{0.73} = 49.32\%$
15. % of silver $= \frac{108}{143.5} \times \frac{7.2}{5.82} \times 100 = 93.1\%$
16. % of Nitrogen $= \frac{14}{1000} \times \frac{29\times\frac{1}{5}}{0.5} \times 100 = 16.24\%$
17. $3.4 = \frac{32}{X} \times 100 \rightarrow X = \frac{32}{3.4} \times 100 = 941$

18. % of P $= \frac{62}{222} \times \frac{0.222}{0.1} \times 100$

19. % of K $= \frac{0.2}{0.5} \times 100 = 40\%$

28. a-r,s; b-r,s; c-p,q; d-p,q

Worksheet – 8

Level – I

1) 3 2) 1 3) 3 4) 2 5) 2 6) 3

Level – II

1) 1	2) 2	3) 2	4) 1	5) 2	6) 2	7) 2	8) 4
9) 3	10) 1	11) 1	12) 1	13) 3	14) 2	15) 3	16) 3
17) 1	18) 3	19) 2, 4	20) 3	21) 2	22) 3		
23) a-s; b-p; c-r; d-q		24) 2	25) 4	26) 3	27) 2		
28) a-r,s; b-r,s; c-p,q; d-p,q							

5. $CH_2O = 12 + 2 + 16$
$= 30.$
$n = \frac{\text{Molecular Mass}}{\text{Empirical Mass}}$
$= \frac{90}{30}$
$= 3$
$= (CH_2O)_3 = C_3H_6O_3$

6. Molecular mass $= 2 \times VD$
$= 2 \times 17$
$= 34$
$n = \frac{\text{Molecular Mass}}{\text{Empirical Mass}}$
$n = \frac{34}{17} = 2 = (HO)_n$
$(HO)_2 = H_2O_2$

7. Molecular Mass $= 2 \times VD$
$= 2 \times 45$
$= 90$
$n = \frac{\text{Molecular Mass}}{\text{Empirical Mass}}$
$= \frac{90}{30} = 3$
$n = (CH_2O) = C_3H_6O_3$

9. $n = \frac{\text{Molecular Mass}}{\text{Empirical Mass}}$
$\frac{78}{13} = 6 = (CH)_n = (CH)_6 = C_6H_6$

10. Empirical Formula $= CH_2O$
Molecular Formula $= (\text{ Empirical Formula })_n$
$n = \frac{M \cdot F.}{E.F.} = \frac{180}{30} = 6$

Molecular Formula = $(CH_2O)_6 = C_6H_{12}O_6$

11. $n = \frac{\text{Molecular Mass}}{\text{Empirical Mass}} = \frac{30}{15} = 2$

$= (CH_3)_2 = C_2H_6$

13. a) $H:O = 2:1$

Carbohydrate weight = 310 g

Carbon residue formed is = 124 g

So from 310 g – 124 g of C removed then the water formed is 186 g

In 186 g of $H_2O \rightarrow H:O = 2:1$

then 18 g of H_2O contains = 2 g of H_2

then 186 g of H_2O contains = 20.66 g of H_2

18 g of H_2O contains = 16 g of O_2

186 g of H_2O contains = 65.33 g. of O_2

$C:H:O$

126:20.66:65.33

$C = \frac{124}{12} = 1.03 = 1$

So Empirical formula is CH_2O

b) 0.0833 mol = 1 g hydrogen

1 mol = x

$x = \frac{1}{0.0833} = 12$ g

In CH_2O – Hydrogen is 2 g $= \frac{12}{2} = 6(CH_2O)_6 = C_6H_{12}O_6$

14. $\%H = \frac{2}{18} \times \frac{1.8}{1.4} \times 100 = 14.2\%$

$\therefore$ % of $C = 100 - 14.2 = 85.8\%$

$\therefore$ Empirical formula = CH_2

15. $C_6H_{14} = \frac{72}{14} = 5.1:1$

16. % of $C = \frac{12}{44} \times \frac{0.458}{0.312} \times 100 = 40\%$

% of $H = \frac{2}{18} \times \frac{0.374}{0.312} \times 100 = 13.3\%$

% of $N = 100 - 53.5 = 46.7\%$

Element	% Composition	Atomic Weight	Relative no. of atoms	Simplest Ratio
C	40	12	$\frac{40}{12} = 3.33$	$\frac{3.33}{3.33} = 1$
H	6.67	1	$\frac{6.67}{1} = 6.67$	$\frac{6.67}{3.33} = 2$
O	53.33	16	$\frac{53.33}{16} = 3.33$	$\frac{3.33}{3.33} = 1$

17. Empirical formula = $CH_4\ N$

Element	% Composition	Atomic Weight	Relative no. of atoms	Simplest Ratio
C	18.5	12	$\frac{18.5}{12} = 1.54$	$\frac{1.54}{1.54} = 1$
H	1.55	1	$\frac{1.55}{1} = 1.55$	$\frac{1.55}{1.54} = 1$
Cl	55.04	35.5	$\frac{55.04}{35.5} = 1.55$	$\frac{1.55}{1.54} = 1$
O	24.81	16	$\frac{24.81}{16} = 1.55$	$\frac{1.55}{1.54} = 1$

18. Empirical formula = $CHClO$

Element	% Composition	Atomic Weight	Relative no. of atoms	Simplest Ratio
C	20	12	$\frac{20}{12} = 1.66$	$\frac{1.66}{1.66} = 1$
H	6.67	1	$\frac{6.67}{1} = 6.67$	$\frac{6.67}{1.66} = 4$
N	46.67	14	$\frac{46.67}{14} = 3.33$	$\frac{3.33}{1.66} = 2$
O	26.66	16	$\frac{26.66}{16} = 1.66$	$\frac{1.66}{1.66} = 1$

19. Empirical formula = $CH_4\ N_2O$
Molecular formula = $(EF)_n = (CH_4\ N_2O)_1 = (NH_2)_2CO$

Element	% Composition	Atomic Weight	Relative no. of atoms	Simplest Ratio

P	43.6	31	$\frac{43.6}{31} = 1.406$	$\frac{1.406}{1.406} = 1$
O	46.4	16	$\frac{26.66}{16} = 3.525$	$\frac{3.525}{1.406} = 2.5$

Empirical formula = P_2O_5

Molecular formula weight = 142

20. A: B: C = 1: 0.5: 1; = $\frac{1}{0.5}:\frac{0.5}{0.5}:\frac{1}{0.5}$; = 2: 1: 2;

Empirical formula A_2BC_2

26. 194 g = 100%

x = 28.9%

x = $\frac{194\times28.9}{100}$ = 56 g of N_2

X = 4 atoms of N_2.

27. MW = $\frac{\text{Weight}}{\text{no.of mole}} = \frac{10}{\frac{1}{4}} = 40$

Ef.wt = 20

$n = \frac{40}{20} = 2 \therefore$ molecular formula = $(HF)_2$

= $(HF)_2$

Worksheet – 9

Level – I

1) A 2) A 3) A 4) B 5) C 6) C 7) A 8) A

5. $2KClO_3 \rightarrow 2KCl + 3O_2$

(2×122.5) g of $KClO_3 \rightarrow (3 \times 22.4)$ lit of O_2

245 g of $KClO_3 \rightarrow$ 67.2 litof O_2

? → 5.6 lit of O_2

$\Rightarrow \frac{5.6 \times 245}{67.2} = 20.4g$

8. Volume = $\frac{20}{2} \times 22.4 = 224$ lit

Level – II

1) D	2) A	3) C	4) B	5) C	6) B	7) B
8) A	9) D	10) C	11) B	12) C	13) A	14) C
15) A	16) C	17) C	18) C	19) A	20) A	21)
22) D	23) A	24) C	25) A	26) A		

27) a-q; b-p,r; c-p,r,s; d-p,r,s 28) D

1. $2KClO_3 \rightarrow 2KCl + 3O_2$

245 g of $KClO_3 \rightarrow$ 67.2 lit of O_2

log of $KClO_3 \rightarrow$?

$\frac{10 \times 67.2}{245} = 2.74\text{lit}$

2. 2 Moles of $Ca_3(PO_4)_2 = 1\text{Mole of } P_4$
   2 Moles $Ca_3(PO_4)_2 = 123$ g of P_4
   1 Mole $Ca_3(PO_4)_2 = x$
   $= \frac{123}{2} = 61.5 = 62$ g
   620 g = 123 g of P_4 of $Ca_3(PO_4)_2$
   1 g = x
   $x = \frac{123}{620} = 0.19 = 0.2$
3. 1 ton = 907184 g
   620 g of $Ca_3(PO_4)_2 = 123$ g of P_4 \
   907184 = x
   $\frac{907184 \times 123}{620}$
   = 179973.6 g
4. $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$
   58.33 g of $Mg(OH)_2 \overset{\text{needed}-2}{\longrightarrow}$ 2 moles of HCl
   1 g of $Mg(OH)_2 \overset{\text{needed}}{\longrightarrow}$?
   $\frac{1 \times 2}{58.33} = 0.034$
5. $2C_8H_{18} + 25O_2 \rightarrow 16O + 18H_2O$
   (2 × 114)g of octane $\overset{\text{required}}{\longrightarrow}$ (25 × 32)g of O_2
   $\Rightarrow \frac{570 \times 800}{228} = 2000$ g of O_2
   228 g of octane required 800 g of O_2
   570 g of octane required?
   $\Rightarrow \frac{570 \times 800}{228} = 2000$ g of O_2
6. $CuO + H_2 \rightarrow Cu + H_2O$
   79.5 g of CuO requires 22.4 lit of H_2 ?
   CuO requires 2.8 lit of H_2
   $\frac{2.8 \times 79.5}{22.4} = 9.93$ g.
7. $CaCO_3 \rightarrow CaO + CO_2$
   100 g of $CaCO_3 \rightarrow 22.4$ lit of CO_2
   20 g$CaCO_3 \rightarrow$?
   $\frac{20 \times 22.4}{100} = 4.48$ lit of CO_2
8. $2CO + O_2 \rightarrow 2CO_2$ (1lit = 1000 cm^3)
   i) 2 Mole of CO → 2 moles CO_2
   100 cm^3 of CO → 100 cm^3 of CO_2
   ii) 2 moles of CO $\overset{\text{requires}}{\longrightarrow}$ 1 mole of O_2
   100Cm^3 of CO → ?
   $\frac{100 \times 1}{2} = 50$ cm^3 of O_2
9. $CaCO_3 \rightarrow CaO + CO_2$
   100 g_{of} $CaCO_3 \rightarrow 56$ g of CaO

200 g of $CaCO_3$ → ?

$\frac{200\times56}{100} = 112$ g of CaO

Purity is 95%

$\therefore \frac{95}{100} \times 112 = 106.4g$

10. $CaCO_3 \rightarrow CaO + CO_{2w}$

100 g of $CaCO_3$ → 50 g of CO_2

12. $Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$

106 g of Na_2CO_3 → 22.4 lit of CO_2

2.12 g of Na_2CO_3 ?

$\frac{2.12 \times 22.4}{106} = 0.4482$

13. $NaHCO_3 + HCl \rightarrow NaCl + CO_2 + H_2O$

84 g of $NaHCO_3$ → 22.4 lit of CO_2

? → 0.56 lit of CO_2

$\frac{0.50 \times 84}{22.4} = 2.1g$

14. $C_{12}H_{22}O_{11} + 12O_2 \rightarrow 12CO_2 + 11H_2O$

1 day = 24hrs

1hr = 34 g of sucrose

$$24 \text{ hrs} = 24\times34=816 \text{ g}$$

$$342 \text{ g of sucrose} \rightarrow (12X32) \text{ g of } O_2$$

$$816 \text{ g of sucrose} \rightarrow ?$$

$$= \frac{816\times12\times32}{342}$$

$$= 916.21 \text{ g}$$

15. $CuO + H_2 \rightarrow H_2O$

79.5 g of Cuo $\overset{\text{requires}}{\longrightarrow}$ 2 g of H_2

0.795 g of Cuo $\overset{\text{requires}}{\longrightarrow}$ x

$\frac{.795\times2}{79.5} = 0.02$ gof H_2

16. $2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$

2 moles of C_2H_2 required 5 moles of O_2

100ml of C_2H_2 required?

$\frac{100\times5}{2} = 250$ml of O_2

17. Weight of pure HCl in 1000 g of 42% acid

$= \frac{42}{100} \times 1000 = 420g$

$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$

98 g of H_2SO_4 → 73 g of HCl

x → 420 g of HCl

$\frac{420 \times 98}{73} = 563.83g$

90% pure H_2SO_4 required is $= \frac{563.83 \times 100}{90}$

= 626.47 g.

18. Volume = $1/2 \times 22.4 + 1/4(22.4)$

= 11.2 + 5.6

= 16.8 lit.

19. $CaO + 3C \rightarrow CaC_2 + CO$

56 g of CaO produced 64 g of CaC_2

The amount of CaO required $= \frac{56 \times 10^6}{64} = 0.875 \times 10^6$ g = 875 kg

Since in the final, sample for every 85 parts of pure CaC_2 the amount of lime present should be 15 parts. Hence 1000 kg of pure CaC_2 will carry $= \frac{100 \times 15}{85}$

= 176.5 kg of CaO

Hence, the total amount of CaO required for 1000 kg of pure CaC_2 = 875 + 176.5 = 1051.5 kg

b. Again, 1000 kg of crade sample will contain $\frac{15 \times 1000}{100}$ or 150 kg of unreacted CaO and 850 kg of pure CaC_2

Also for the preparation of 850 kg of pure CaC_2, amount of CaO required

$= \frac{56 \times 850}{64} = 743.75$ kg

Hence, the total quality of CaO required for 1000 kg of crude sample

= 743.75 + 150

= 893.75 kg

20. $\underset{28g}{N_2} + \underset{6g}{3H_2} \rightarrow \underset{34g}{2NH_3}$

20 kg of N_2 requires 4.2 kg of Hydrogen but in the reaction Hydrogen is only 3 kg so, limiting reagent is H_2.

wt. of NH_3 obtained from $6KgNH_2 \rightarrow 34$ kg

3 kg of H_2 → x

$\frac{3 \times 34}{6} = 17$ kg

21. $2H_2 + O_2 \rightarrow 2H_2O$

10 g 64 g

5 mole 2 moles

2 Moles H_2 → 1 Mole of O_2

5 Mole H_2 → 2.5 Mole of O_2

Here O_2 is limiting reagent

22. $C + O_2 \rightarrow CO_2$

12 g22.4 L

The amount of Carbon unreacted = $40 \times \frac{10}{100}$ g = 4 g

So, amount of Carbon reacted = $(40 - 4)$g = 36 g

At, STP, for the combustion of 12 g of C, Oxygen required = 22.4 L

∴ for the combition of 36 g of C, Oxygen required will be $= \frac{22.4}{12} \times 36 = 67.2$ L

23. The number of electrons present in 18ml of water is

1000ml of water = 1000gm of water
18ml of water = 18 g of water
So 1 atom of water comtains 10 electrons
So $10 \times 6.023 \times 10^{23}$
6.023×10^{24} electrons.

24. $2KClO_3 \rightarrow 2KCl + 3O_2$
$245\ g \rightarrow 149\ g$ of KCl
$24.5g \rightarrow ?$
$\frac{24.5 \times 149}{245} = 14.9$ gKCl
$245\ g_{of}\ KClO_3 \rightarrow 3 \times 22.4$ litof O_2
$24.5\ gKClO_3 \rightarrow ?$
$\frac{24.5 \times 3 \times 22.4}{245} = 6.72$ lit of O_2
$V_1 = 6.72$ lit $V_2 = ?$
$T_1 = 273k\ T_2 = 300k$
$P_1 = 760$ mm
$P_2 = 750$ mm
$V_2 = \frac{P_1 V_1}{T_1} \times \frac{T_2}{P_2}$
$= \frac{760 \times 6.72}{273} \times \frac{300}{750}$
$= 7.48$ lit.

25. weight of other component $= \frac{2.33 \times 10^{-23}}{1.66 \times 10^{-24}} = 14.04a \cdot m \cdot u.$
total wt. ofmolecule $\Rightarrow C \rightarrow 9 \times 12 = 108.$
$H \rightarrow 13 \times 1 = 13$
other component = 14.04
$\therefore 108 + 13 + 14.04 = 135.04$

26. (i) $Mg + 2HCl \rightarrow MgCl_2 + H_2 \uparrow$
$MgCl_2 = 95\ g = 1$ mole
$9.5\ g = 0.1$ mole
(ii) 1 Mole $Mg \rightarrow 1MgCl_2$

28. Na_2O: Na $\rightarrow 2 \times 23$
$O \rightarrow 1 \times 16$
total = 62,1 mole $\rightarrow 62$
$? \rightarrow 248$
$= \frac{248}{62} = 4$ moles

Worksheet – 10
Level – I

1. B	4. B	7. C	10. D
2. A	5. B	8. A	
3. A	6. A	9. A	

6. C: O $\Rightarrow$ 12: 32

Level – II

1) 4	2) 3	3) 3	4) 1	5) 1	6) 1	7) 3	8) 1
9) 1	10) 2	11) 1	12) 3	13) 3	14) 1,2	15) 2	16) 1
17) 2	18) a-s; b-r; c-q; d-p		19) 1, 3, 4	20) 4	21) 4	22) 2	
23) a-s; b-q; c-p; d-s		24) 2	25) 1	26) 4	27) 24		
28) a-q,r,s,t ; b-p; c-r,s ; d-p							

12. According to law of conservation of mass
mass of reactants = mass of products
∴ mass of reactants (or) products
$= 5.3 + 6.0$
$= 11.3$gr

13. $2CuO \rightarrow 2Cu + O_2$
According to conservation of mass
$3.672 = 2.938 + W_2$
$WO_2 = 3.672 - 2.938 = 0.734$ g
$Wcu : W_2 = 2.938 : 0.734$
$= 4 : 1$